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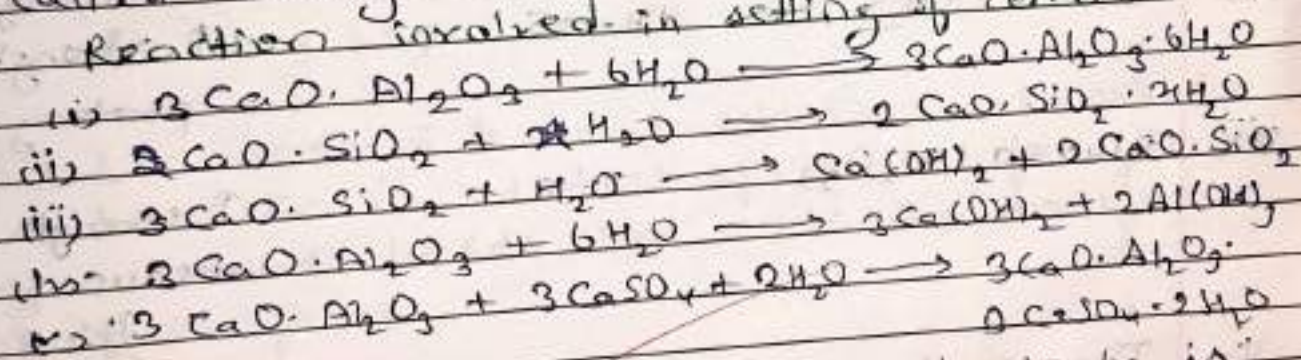
Assignment - 4

Unit 4

Q1 What is ~~fast~~ setting cement?

Ans When mixed with water, cement sets to a hard mass. It first forms a plastic mass which hardens after some time due to 3-dimensional cross-links between the $-Si-O-Si-$ & $-Si-O-Al-$ chains. The first setting occurs within 24 hours whereas the subsequent hardening requires a fortnight when it covered by a layer of water. This transition from plastic to solid state is called setting.

Reaction involved in setting of cement.



The fast-setting tricalcium aluminate is removed to slow down the setting process. A quick setting will give rise to crystalline hydrated calcium aluminate. A slower setting yields the colloidal gel that imparts water strength to the set mass. Thus gypsum helps in ~~equat~~ regulating the setting time of cement.

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Q2 What should be the flash point of good lubricant?

Ans The flash point of a volatile material is the lowest temperature at which vapours of the material will ignite for a moment when an ignition source brought near to it. The lubricating oil should have flash point minimum above its working temp. It should be high.

Q3 Write chemical composition of constituents of cement.

Ans	Constituent	Range by mass	Function
i	Lime	60-70	↑ strength
ii	Silica (SiO_2)	17-25	↑ strength & prolong setting
iii	Alumina (Al_2O_3)	3-8	↑ strength but reduces setting time.
iv	Iron-oxide (Fe_2O_3)	2-4	Impart grey colour.
v	MgO	1-5	↑ strength & hardness contribute to soundness
vi	SO_2 (Sulphuric oxide)	1-3	Imparts soundness if present in small amt.

Q.iii Alkali oxide $\text{Na}_2\text{O} / \text{K}_2\text{O}$ 0.2-1.5 Causes eff.

Q4 What is glass wool & write its uses?
 Ans Fibrous wool like material composed of mixture of fine filament of glass. These are made by action of steam jet on creeping molten glass down from very fine holes.

Properties

- (a) Non combustible & fire proof
- (b) Poor conductor of electricity
- (c) Heat proof.
- (d) Resistance to most chemicals
- (e) Very high tensile strength.

Uses:- Electrical & sound insulation, heat insulation, indomestic & industrial appliances, for filtration of corrosive liquids like acids. In manufacture of glass fibre.

Q5 Write chemistry of setting & hardening of cement.

Ans Setting:- When cement is mixed with water & allowed to stand, it sets into a hard & rigid mass due to chemical reaction & gel formation.

Hardening:- Converting into compact rock like material & developing of strength due to crystallisation is known as hardening.

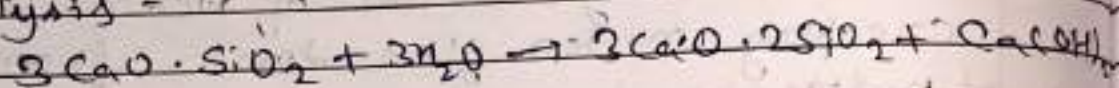
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The process of setting & hardening of cement are believed to be partly a chemical & partly a physical changes.

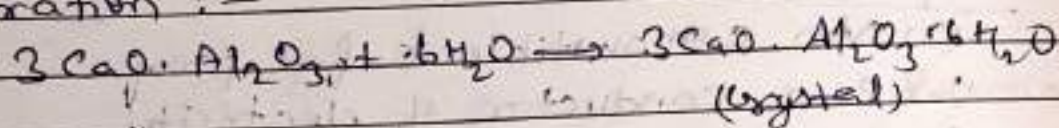
Chemical change - Hydration

Physical change - Formation

Hydration :-



Hydration :-



Q.6

Ans

Write a short note on annealing of glass.

Annealing is a crucial heat treatment process used in glass manufacturing to relieve internal stresses that develop during the cooling & shaping of glass. When hot glass is cooled too quickly, uneven temp. distribution can cause stress, making the glass brittle & prone to cracking or breaking.

In the annealing process, the glass is slowly cooled in a controlled environment, typically in an annealing oven or lehr. The temp is maintained just below the glass's softening point and gradually reduced over time. This allows the internal stresses to relax & ensures uniform temp distribution throughout the material.

hot glass article

300°C	800°C	700°C	600°C	500°C	400°C	300°C	200°C
→	→	→	→	→	→	→	→

Anneal,
1 hr

Q7 Explain extreme-pressure lubrication?
 When the machine runs under the conditions of high loads the temp. rises between the moving surface of the metals. Thus, under such severe conditions ordinary fluid film or boundary fluid film is not effective because it may vaporize or fail to stick on the metal surfaces.

Properties of Lubricant:-

1. Viscosity
2. Ash content
3. Acid value
4. Carbon residue
5. Oxidizability
6. Volatility
7. Saponification
8. Flash & fire point
9. Aniline point
10. Cloud & pour point

Q8 Write structure of graphite & MoS₂ why are these used as solid lubricants?

Graphite - a molybdenum disulphate (MoS₂)
 Graphite is made of carbon atoms in hexagonal layers.
 Strong bonds within layers, weak forces between layers.
 layers slide easily - reduces friction.

MoS₂ (Molybdenum Disulfide) :-

- Mo atom sandwiched between two sulfur layers.
- Strong Mo-S bonds in layer, weak forces between layers.
- layers slide easily - acts as a lubricant.

Use - layered structure allows easy sliding
 reduce friction & wear.
 Work in dry, high-temp or vacuum condition.

Q9 Describe manufacturing of cement by Rotary kiln Process.

Ans Raw material - limestone (CaCO_3), clay (Alumina, silica) and other additives.
 crushed & mixed in correct proportions.

Raw material Preparation:-

Grinding - The mixture is ground into a fine powder (called raw meal).

Rotary kiln Process:-

Raw meal is fed into a rotary kiln (a long, rotating cylinder slightly inclined).
 Three main zones.

(i) Drying zone - Removes moisture from raw meal using hot gases.

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2. **Heating (Calcination) Zone:**
 Temperature - $800-1000^{\circ}\text{C}$
 Limestone decomposes -
 $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$

3. **Burning (Clinkering) Zone**
 Temperature - $1400^{\circ}\text{C} - 1500^{\circ}\text{C}$
 material react to form clinker (nodules of Calcium silicates etc.)

4. **Cooling** Clinker is cooled using air.
 Sudden cooling is known as annealing.

5. **Grinding**:- Clinkers is ground with gypsum to form cement.

6. **Packaging**:- Final cement is packed & transported.

